Stat 1040:

Homework Solutions

Stat 1040
Homework 1 Solutions

Chapter 2

1. The statement is false - the data do not show that if you have to fly, it is safer to do so on a commuter airline. We cannot compare the numbers given - we need to compare rates. To decide what the data DO show, we need to know how many people flew on commuter airlines versus scheduled carriers, and then we can calculate the rates and compare.

4. (a) They studied the groups separately to eliminate the effects of the confounding factors of age and gender.

   (b) That is not an appropriate conclusion because there are confounding factors. For example, those who recently stopped smoking may have done so on doctor's orders, because they had severe health problems.

5. Zinc sulfate should not be given to treat this disease because the double-blind experiment shows no effect. It seems likely that the apparent effect for the single-blind study is caused by either physician bias or the possibility that the physicians broke the blind by communicating (perhaps unintentionally) the patients' treatment status with them.

7. (a) This is an observational study - the patients chose to be on oral contraceptives.

   (b) Age, education, and marital status are variables that are known to have associations with cirvical cancer and pill use. They adjusted for these variables to try to eliminate problems due to these confounding factors.

   (c) They are likely to be more sexually active than non-users. This is related to marital status, but marital status doesn't tell the whole story.

   (d) The conclusions are not justified by the study. There are several possible confounding factors that were not adjusted for. For example, the number of sexual partners is likely to be higher of pill-users than non-pill users, and it could be that cirvical cancer is a sexually transmitted disease.

9. (a) The experiments did not confirm the results of the observational studies - the observational studies suggested that fresh fruits and vegetables might lower rates of colon and lung cancer, whereas the randomized, controlled experiments found no relationship and the opposite relationship for colon cancer and lung cancer respectively.

   (b) This is true - for example, the people who eat lots of fruit and vegetables probably exercise more, they may be wealthier, they may be younger, all of which may be associated with lower colon and lung cancer rates.

   (c) This is false - the experiments were randomized - there should be no confounding factors.

10. (a) This was an observational study - mothers get to choose their behavior.

    (b) Yes, there was an association - fatter children tended to have more "controlling" mothers.

    (c) That would explain the association, but it is not the only possible explanation.

    (d) A gene would not explain the association unless the gene were also linked to "controlling" behavior of the mothers, which is conceivable but seems unlikely.

(e) Perhaps when mothers see their children getting fat, they become more "controlling". This is a causal relationship but in the opposite sense of the one on part (c). Or, there could be other confounding factors. For example, perhaps "type A" mothers have some genetic factor that leads to their children being fat and also causes the mothers to be "controlling.

(f) No, the data do not support the advice, because there is insufficient evidence of a causal relationship.

11. (a) Treatment: those prisoners who complete boot camp. Control: everyone else, including those who start but do not complete boot camp, and those who don't even try it.

    (b) Observational study.

    (c) False. The data found an association. But there are certainly confounding factors, not the least of which is the psychological makeup of those prisoners who complete boot camp.

12. False. Democrats could be concentrated in wards with low turnout, as in the following example (Simpson's paradox in all its glory).

	Democrats		Republicans	
	Registered	Turnout	Registered	Turnout
Ward A	1000	100	100	5
Ward B	100	60	1000	500
Total	1100	160	1100	505
		(14.5%)		(45.9%)

Stat 1040
Homework 2 Solutions

Chapter 3

2. (a) There are more children age 1, because the histogram is higher there.
   (b) There are more at 31.
   (c) There are more age 35-44, that block has larger area.
   (d) 50%

4. (a) 25%
   (b) 99%
   (c) 140-150 mm
   (d) 135-140 mm
   (e) About 10.5% (= 5 * 2.1%)
   (f) 102-103 mm
   (g) One of the millimeters between 115 and 120, say 117–118.

6. Lists (i) and (ii) but not (iii).

10. (a) Picture attached.
    (b) In 1880, people did not know their ages accurately and probably rounded off.
    (c) In 1970, however, people knew better when they were born.
    (d) In 1880 there seemed to be a stronger preference for even digits (except for the ever-popular 5). This is probably due to rounding. In 1970 the preference was weak.

Chapter 4

1. (a) Average=50, SD=5.
   (b) 48, 50, 50 are within 0.5 SDs of average, that is, they fall in the range 47.5-52.5. 48,50,50,54,57 are within 1.5 SDs of the average, that is, in the range 42.5-57.5.

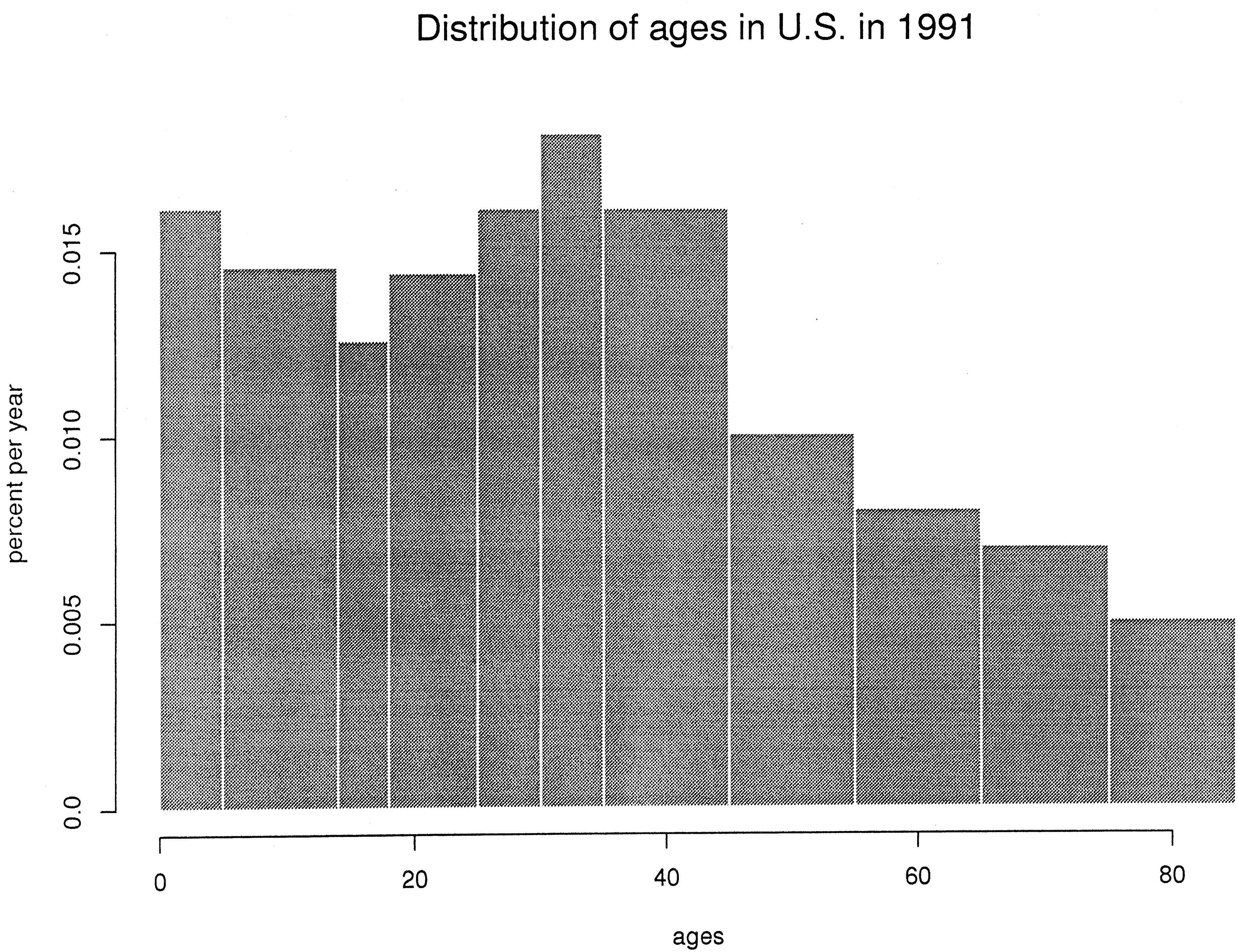
3. (a) 5, because only three of the numbers are smaller than 1, and none are bigger than 10.
   (b) 3, because if the SD is 1, then the entries 0.6 and 9.9 are much too far away from the average; the SD can't be 6 because none of the numbers are more than 6 away from the average.

6. (a) (i) 60 (ii) 50 (iii) 40
   (b) (i) median is bigger than average – long left hand tail (ii) median is about equal to the average – symmetry (iii) median is less than the average – long right had tail
   (c) 15, because most of the area is within 50 of the average, so 50 is too big; and only a small portion of the area is within 5 of the average, so 5 is too small.
   (d) False. The two histograms are almost mirror images and have about the same SD. (Remember the rule for what happens to the SD when changing the sign of a list of numbers.)

7. (a) Average weight of men = 66×2.2 = (about) 145 pounds. SD = (about) 20 pounds. Average weight of women = (about) 121 pounds. SD = (about) 20 pounds.

   (b) 68%. Then range is from one SD below the average to one SD above.

   (c) Bigger than 9kg. When you combine the men and women together, the spread increases.


Distribution of ages in U.S. in 1991
percent per year
0.0
0.005
0.010
0.015
0
20
40
60
80
ages

5

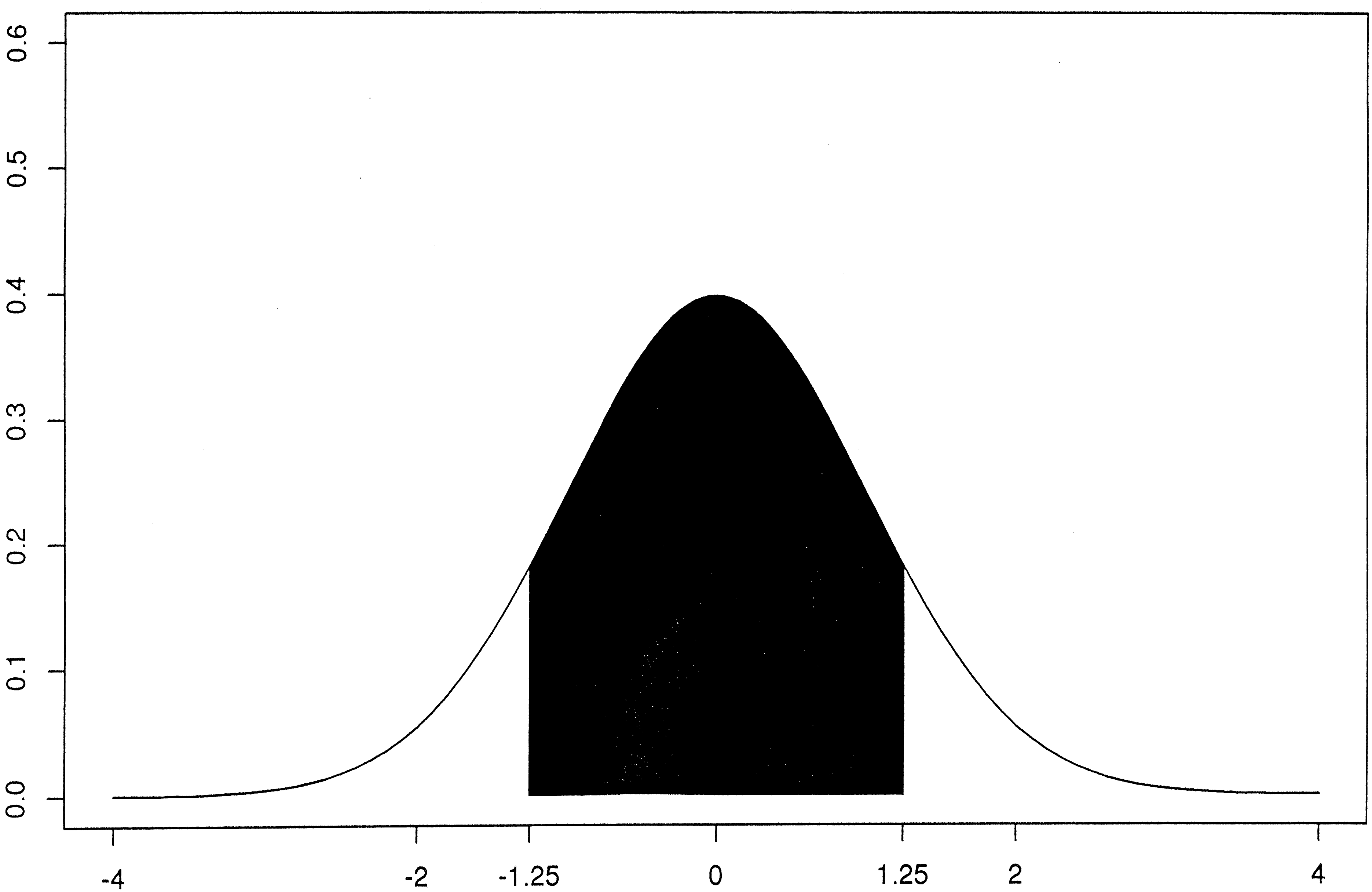
Stat 1040
Homework 3 Solutions

Chapter 5

1. (a) The Normal Table in the back of the book shows that about 79% of the area falls between -1.25 and 1.25. For 25 test scores, 79% = (about) 20.
   (b) 18 scores were actually within 1.25 SDs of the average, or within the range (37.5, 62.5).
3. (a) In 1967, a score of 600 was about 1.22 SDs above the average, and the percentage scoring over 600 was about 11%.
   (b) In 1994, a score of 600 was about 1.61 SDs above the average, and the percentage scoring over 600 was about 5%.
4. (a) 20%
   (b) 12%
5. The percentage of men with heights between 66 inches and 72 inches is exactly equal to the area between 66 inches and 72 inches under the histogram. This percentage is approximately equal to the area between -1 and 1 under the normal curve.
7. (a) 350 is 1.5 SDs below the average. This student is at the 7th percentile of the score distribution because under the normal curve, 7% of the area lies below -1.5.
   (b) About 75% of the area under the normal curve lies below 0.7 SDs, so a student needs a score of about 570 on the Math SAT to be at this percentile.
8. (a) True
   (b) False. See page 92. Adding a constant to the data does not change the spread.
   (c) True
   (d) True. All deviations from the average are doubled.
   (e) True
   (f) False. The deviations from the average will also have their signs changed, but negative go away in the squaring. Besides, SD is never negative.
9. (a) False. The list 10, 11, 60 has a median of 11 but an average of 27.
   (b) False again. For the list 10, 11, 60, two-thirds of the entries are below the average.
   (c) False. Income data doesn't follow the normal curve even with large samples, because incomes aren't measured below 0. See page 88.
   (d) False. If they both follow the normal curve, then you might get about 68% of the area between 40 and 60 for both lists. But that is still only an approximation, and there are many other possibilities.

Chapter 6

4. About 0.03 inches.



7

Stat 201
Homework 4 Solutions

Chapter 6 - Special Review Exercises

6. (a) $(600 + 650)/2 = 625$.
   (b) It would be more than 125 because the distributions are different, so when they are combined there will be more variability.

8. It would be too low, because the curve is lower than the histogram in that range.

10. False. The data are cross-sectional. One possible explanation is that when those 70 year olds were young, people were forced to write with their right hand, now they are not. To decide what happens to people as they age, you need a longitudinal study.

13. (a) If the drivers were older, they might just have higher rates heart disease because of age. (Older people have higher rates of heart disease).
    (b) Any effects of exercise might take a while to show up.
    (c) Drivers are more likely to be similar to conductors with respect to age, education, income, socioeconomic background, ethnicity, etc.
    (d) This is an observational study and there are several plausible confounding factors. For example, drivers might have been heavier to begin with.
    (e) See if the drivers were heavier (had larger uniforms) than the conductors at the time of hire. If both groups were the same, initial weight is not the confounding factor. If drivers were heavier to begin with, the argument that exercise matters is much weaker. Note: looking at size after 10 years does not help us to decide.

Chapter 8

1. d. (a has the wrong means, b has the wrong SD's, and c has too much correlation).

2. (a) Negative: older cars are less fuel-efficient.
   (b) Richer people tend to own newer cars and maintain them better.

4. 0.3: (It should be positive, but not as high as .9!)

11. It is -1. (Sketch a scatter diagram if you have trouble seeing this).

Stat 1040
Homework 5 Solutions

Chapter 9

2. (a) False - it's the other way round. Below-average values of the dependent variable tend to be associated with above-average values of the independent variable.
   (b) False. For example, height in inches is usually less than weight in pounds, but the correlation coefficient is positive.

8. False - these are cross-sectional data not longitudinal data. Younger people were born more recently and education levels have been going up over time.

11. Quite a bit less because this is an ecological correlation.

12. (a) Education is measured in whole years.
    (b) Some dots correspond to more than one couple.
    (c) A (iv), B (iii), C (i).

Chapter 10

1. A (i), B (iii), C (ii).

2. (a) 112.
   (b) 112.

3. (a) 64 inches.
   (b) 62 inches.
   (c) 63 inches.
   (d) 63 inches.

4. (a) 15 years.
   (b) 13.5 years.
   (c) This is just the regression effect.

Stat 1040
Homework 6 Solutions

Chapter 11

1. (v).

2. Something is wrong - GPA's are between 0 and 4. Even if you always predicted 2, you would have an rms error of less than 3.12.

3. (a) $\sqrt{1-.8^2} \times 2.5 = 1.5$ inches.

   (b) $\sqrt{1-.8^2} \times 1.7 = 1.0$ inches.

4. (a) $\sqrt{1-.6^2} \times 15 = 12$.

   (b) This student is 1.2 SD's above average on the midterm, and should be $.6 \times 1.2$ SD's above average on the final. That's 10.8 points above 55, i.e. 65.8.

   (c) $\sqrt{1-.6^2} \times 15 = 12$.

5. (a) The area to the right of 80 is the area to the right of 1.67 under the standard normal curve, which is about 5%.

   (b) New average = 65.8, new SD = 12. Standard units = 1.18. Area = 12%.

6. The conclusion seems as though it should be correct, but it does not follow from the data because there could be confounding factors. For example, it could be that better students just spend more time on homework.

7. Option (iii) is correct because of the regression effect. (The students who scored 75 on the math test should average a bit less than 75 on the physics test, so less than half of them will score over 75.)

12. Men with 18 years of education should have blood pressure of about 122 mm, on average. (Using the regression method). So, his blood pressure was slightly high.

Stat 1040
Homework 7 Solutions

Chapter 12

2. income $= 960\times$ height $-37,400$. It seems that taller men make more money, on average.

3. The slope means that the heavier men are taller, on average, than the lighter ones. For example, the 141-pound men average about 0.047 inches taller than the 140-pound men.

5. False - there are two lines, you can't just solve the equation. (The real line has slope 0.32).

8. (a) True
   (b) False
   (c) False
   (d) True
   (e) True

9. 1/2000.

10. No, it's likely to be too high. You need to use the other line - a better estimate is around $26,000.

11. Something's wrong because the line should go through the point of averages.

12. No - there is no consistent pattern.

Stat 1040
Homework 8 Solutions

Chapter 13

2. Option (i) has higher chances and will win you more money.

3. Option (ii) is better because the un-ordered version has higher chances. (24 times as high.)

6. (a) True.
   (b) True.
   (c) False. The correct chance is (1 / 52) * (1 / 51) because these events are not independent.

7. (c) is correct. Every possible string of H's and T's is equally likely.

8. (a) (4/6)*(4/6)*(4/6)*(4/6) = (about) 0.20 or 20%.
   (b) (2/6)*(2/6)*(2/6)*(2/6) = (about) 0.012 or 1.2%.
   (c) This is the opposite of (a), so the chance is 100% - 20% = 80%.

9. (a) $(1/6)^{10}$ = 1/60,466,176.
   (b) 1 - 1/60,466,176 = (about) 1.
   (c) $(5/6)^{10}$ = (about) 0.16 or 16%.

11. Option (ii) is better. You have the same 50-50 chance of winning one dollar, but for option (ii) you have no way to lose money.

12. (3/100) * (2/99) * (1/98) = (about) 6/1,000,000.

Stat 1040
Homework 9 Solutions

Chapter 14

1. (a) (1/6)*(1/6) = 1/36.

   (b) (3/36) = (1/6). (Use the chart of possible rolls.)

4. Option (i) is better, because if the first card isn't a queen, you still have a shot at it on the second card.

6. "If you want to find the chance that at least one of the two events will happen, check to see if they are mutually exclusive. If so, you can add the chances."

   "If you want to find the chance that both will happen, check to see if they are independent. If so, you can multiply the chances."

7. 1 - $(3/5)^4$ = (about) .87.

8. 1. You can't avoid drawing a two.

9. List out the possible draws in a chart like this:

		Box B			
		1	2	3	4
	1	11	12	13	14
Box A	2	21	22	23	24
	3	31	32	33	34

   (a) 3/12.

   (b) 3/12.

   (c) 6/12.

10. Option (ii) is better, giving you 3 chances in 6 to win (each time). Option (i) only gives you 2 chances in 6 to win (each time).

11. (a) (13/52)*(12/51)*(11/50) = (about) .01.

    (b) (39/52)*(38/51)*(37/50) = (about) .41.

    (c) 1 - (13/52)*(12/51)*(11/50) = .99. (This is the opposite of (a).)

Stat 1040
Homework 10 Solutions

Chapter 15

3. To have more girls than boys, they must have either 3 or 4 girls. The chance of 3 girls is

$$\binom{4}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^1 = 4 \times .0625 = .25$$

and the chance of 4 girls is

$$\binom{4}{4}\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right)^0 = \left(\frac{1}{2}\right)^4 = .0625,$$

so the chance of getting more girls than boys is $.25 + .0625 = .3125$.

4. This is false because the draws are made without replacement so the draws are dependent.

8. The chance of getting exactly 2 heads amongst the first 5 tosses is

$$\binom{5}{2}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^3 = \frac{5 \times 4}{2} \times .03125 = .3125.$$

The chance of getting exactly 4 heads amongst the last 5 tosses is

$$\binom{5}{4}\left(\frac{1}{2}\right)^4\left(\frac{1}{2}\right)^1 = 5 \times .03125 = .15625.$$

So the chance that these two things both happen is $.3125 \times .15625 = .0488$ (we can multiply because they are independent).

9. (a) (i) by the multiplication rule.
   (b) (viii) the chance is zero because they can't both be the king of spades.
   (c) (iv) by the addition rule (these events are mutually exclusive).
   (d) (viii) because these events are not mutually exclusive.
   (e) (vii) like (a), but twice as likely.

11. (a) The chance of having 17 pairs where the smoker dies first is:

$$\binom{22}{17}\left(\frac{1}{2}\right)^{17}\left(\frac{1}{2}\right)^5 = 26334 * \left(\frac{1}{2}\right)^{22}.$$

The chance of having 18 pairs where the smoker dies first is:

$$\binom{22}{18}\left(\frac{1}{2}\right)^{18}\left(\frac{1}{2}\right)^4 = 7315 * \left(\frac{1}{2}\right)^{22}.$$

The chance of having 19 pairs where the smoker dies first is:

$$\binom{22}{19}\left(\frac{1}{2}\right)^{19}\left(\frac{1}{2}\right)^3 = 1540 * \left(\frac{1}{2}\right)^{22}.$$

The chance of having 20 pairs where the smoker dies first is:

$$\binom{22}{20}\left(\frac{1}{2}\right)^{20}\left(\frac{1}{2}\right)^{2}=231*\left(\frac{1}{2}\right)^{22}.$$

The chance of having 21 pairs where the smoker dies first is:

$$\binom{22}{21}\left(\frac{1}{2}\right)^{21}\left(\frac{1}{2}\right)^{1}=22*\left(\frac{1}{2}\right)^{22}.$$

The chance of having 22 pairs where the smoker dies first is:

$$\binom{22}{22}\left(\frac{1}{2}\right)^{22}\left(\frac{1}{2}\right)^{0}=\left(\frac{1}{2}\right)^{22}.$$

Adding all these gives a chance of approximately 0.8 of 1%. This is saying that if each twin in the pair were equally likely to die first, there would be almost no chance that we would observe as many as 17 smokers dying first. We conclude that 17 is too many to be due to chance error.

(b) Using calculations like those in (a), the chance is about 0.2 of 1%.

(c) Using calculations like those in (a), the chance is about 25%.

(d) i. Only for lung cancer.
   ii. No - these twins were genetically identical.
   iii. Yes, this is a possible explanation in all cases.

Chapter 15 - Special Review Exercises

1. False because the population size increased too. We must compare rates. It turns out that the population size was 203 million in 1970 and 249 million in 1990, so the murder rate actually decreased slightly.

3. False - it's like the Berkeley admissions data. The pooled percentage of women admitted is

$$\frac{10+18}{79+102}\times 100\% = 15.5\%$$

and for men it's

$$\frac{250+331}{1312+1259}\times 100\% = 22.6\%,$$

so the pooled selection ratio is 68.6%.

4. (a) is correct. (b) has too much area and (c) has the wrong units.

5. About 1 year. A month would be too little variability, and 5 years would be too much.

6. False - these are cross-sectional data so we can not say anything about what happens to a group of people over time. Perhaps what accounts for the pattern is that the older people were less educated so they tended to be employed in less well-paid jobs.

9. (a) The 70th percentile is .52 in standard units, so this one scored $50+(0.52\times 20) = 60$ points. The 80th percentile is .84 in standard units, so this one scored $50 + (0.84 \times 20) = 67$ points. The brothers differed by around 7 points.

(b) About 9 points.

12. (a) True.
    (b) True.
    (c) False.
    (d) False.

Stat 1040
Homework 11 Solutions

Chapter 16

1. Option (iii) is the best because the chance error is not likely to be exactly zero, but it is likely to be small compared to the number of draws.

3. Both are wrong. Luck and the law of averages are irrelevant because each spin is independent of those that go before. The chances stay the same.

4. (a) 60 rolls. To win, you need a large percentage error, and that is more likely in 60 rolls.
   (b) 600 rolls - now you want a small percentage error.
   (c) 600 rolls again - you want a small percentage error.
   (d) 60 rolls because to get exactly the expected value means getting exactly zero chance error, and that is more likely with fewer rolls.

6. Possibility (i) is more likely because you want the percentage error to be at least 66 2/3% - 50% = 16 2/3%. The percentage error is more likely to be at least 16 2/3% in the short run.

7. The score is like the sum of 25 draws from a box with one ticket marked "4" and 4 tickets marked "-1".

8. The net gain is like the sum of 50 draws from a box with 4 tickets marked "$8" and 34 tickets marked "-$1".

9. Option (ii) is the best. Say the percentage of reds in the box is 60%. Then to win, we want the percentage error to be small, so that the actual percentage will be greater than 50%. This is more likely in the long run.

10. (a) 30/200 = 0.15.
    (b) -20/200 = -0.1.
    (c) average = sum/200.
    (d) They are the same because 5/200 = .025.

Stat 1040
Homework 12 Solutions

Chapter 17

4. The number of aces in 180 rolls of a fair die is like the sum of 180 draws from a box with 1 ticket marked "1" and 5 tickets marked "0". The average of the box is 0.1667 and the SD of the box is 0.3727. So the expected value of the sum is $180\times 0.1667 = 30$, and the standard error of the sum is $\sqrt{180} \times 0.3727 = 5$. In other words, we expect around 30 aces, give or take 5. The range 15 to 45 is 3 SE's each side of the EV, so we expect 99.7% of the people to get values in that range.

7. (a) $321/100 = 3.21$.
   (b) $3.78\times 100 = 378$.
   (c) The average will be between 3 and 4 if the sum is between 300 and 400. The average of the box is 3.5, and the SD is 1.7, so the expected value of the sum is 350 and the standard error is 17. The chance is about 99.7%.

9. (a) is false and (b) and (c) are true. For case (i), the box has 12 tickets marked "2" and 26 tickets marked "-1". The average of the box is -\$0.053 and the SD is \$1.39. The expected value for the sum is -\$53, and the standard error is \$44. For case (ii), the box has 1 ticket marked "35" and 37 tickets marked "-1". The average of the box is -\$0.053 and the SD is \$5.76. The expected value for the sum is -\$53, and the standard error is \$182. Because the standard error is larger for (ii), the chance of coming out ahead is larger for (ii). So is the chance of either winning or losing more than \$100. (You can check out the numbers using the normal curve).

11. The box has 3 tickets marked "0", one ticket marked "1" and one ticket marked "3". The average of the box is 0.8 and the SD is 1.2, so the expected value of the sum is 80 and the standard error is 12. The sum will be around 80, give or take around 12 or so.

Chapter 18

2. (a) The average of the box is 4 and the SD is 2.24. The expected value for the sum is 1600 and the standard error is 45. The standard units is -2.22 and the area is about 99%.
   (b) The number of "3"s is like the sum of 400 draws from a box with 3 tickets marked "0" and 1 ticket marked "1". The average of the box is .25 and the SD is 0.433. The expected value for the sum is 100 and the standard error is 8.66. The standard units is -1.15 and the area is about 12%.

4. The box has one "0" and one "1". The average of the box is 0.5 and the SD is 0.5. The expected value of the sum is 12.5 and the standard error is 2.5. The standard units is $0.5/2.5=0.2$ and the area is approximately 15.54%.

6. Something is wrong with COIN. The average of the box is 0.5 and the SD is 0.5. The expected value of the sum is 500,000 and the SE is 500. COIN is more than 4 SE's away from the expected value, and that is too many SE's.

9. (a) is true, because the average of the box is 0.01 and the SD is .0995, so the expected value is 1 and the SE is (approximately) 1. (b) is false because we cannot use the normal curve - the box has too many 0's.

Stat 1040
Homework 13 Solutions

Chapter 19

5. Too high. Larger households are more likely to have someone home, so it's like the sample is substituting larger households for smaller ones.

6. No. The hallmark of a probability sample is the ability of the investigators to compute the chance that any particular individuals in the population will be selected for the sample. See page 354. In this sample, how can the investigator measure things such as the chance that a certain student will pass by at a certain time?

9. The hospital with fewer live male births is more likely to have 55% or more male births. This question is much like numbers 4 and 6 from chapter 16. The SE as a percentage decreases as the number of draws increases, so the hospital with more live births will have a percentage of male births closer to the expected value of 52%, and a smaller chance of having a percentage above 55%.

12. (a) The survey isn't representative of all teenagers because only "high achievers" were surveyed.

    (b) Probably not. Important questions we should ask: How were the 5,000 chosen? With only 1,957 completed, returned surveys what about non-response bias?

Chapter 20

3. (a) 50,000.

   (b) Each ticket in the box shows a zero (for each form with gross income less than or equal to \$50,000) or a one (for those forms with gross income over \$50,000).

   (c) False. The SD is actually sqrt(0.20 * 0.80) = 0.4.

   (d) True.

   (e) The number of audited forms in the sample with income over \$50,000 is like the sum of 900 draws from the box described in 3.b. The EV for this box is 180, and the SE is sqrt(900) * SE of the box = 12. As a percentage of the number of draws, the EV is 20% and the SE is 1.3%. This means that we expect the number of audited forms in the sample with income over \$50,000 to be 20%, give or take 1.3% or so. The chance that the percentage of the sample falls between 19% and 21% is equivalent to finding the area under the normal curve from -0.75 to +0.75 which is about 55%.

   (f) We have no way of calculating this chance. We need to know the percentage of forms that have gross income over \$75,000 in order to find an expected value and standard error. Besides all that, the data will probably not be normal, and our methods won't be valid anyway.

4. The box we'll use in this question does not have zeroes and ones in it. Sampling 900 tax forms for the total gross income of all 50,000 forms is like finding the sum of 900 draws from a new box which has a gross income on each ticket – one ticket for each tax form – making 50,000 tickets in all. We know from the information given in the problem that the average of this box is \$37,000 and the SD is \$20,000.

(a) 50,000 again.

(b) a gross income.

(c) True.

(d) True.

(e) The chance that the total gross income is over \$30,000,000 is the same as the chance that our sample sum is over \$30,000,000. We expect our sample sum to be (900)*(\$37,000) = \$33,300,000. The SE of our sample sum is sqrt(900)*(\$20,000)=\$600,000. So the chance of our sample sum being over \$30,000,000 is equivalent to the area under the normal curve to the right of -0.5 which is about 70%.

6. Statement (ii) is correct. The accuracy of California's sample will be higher because California's sample will be larger.

7. Statements (a), (c) and (e) are true, while (b), (d) and (f) are false. For (b) and (f), we know exactly the expected value for the percentage of 1's among the draws, and we know exactly the percentage of one's in the population. We only attach measures of chance error to quantities we don't know, such as the percentage of one's that come up in a sample we happen to take. That amount will be different for each sample, whereas the values mentioned in (b) and (f) don't change from sample to sample.

Stat 1040
Homework 14 Solutions

Chapter 21

1. The sample is like 500 draws from a box with 25,000 tickets. The tickets are marked 0 (does not have a computer) or 1 (has a computer). We do not know how many tickets are marked 0 and how many are marked 1. We can approximate the proportion of 1's using the sample percentage: $(79/500)\times$ 100% $= 15.8\%$. We can approximate the SD of the box by $\sqrt{.158 * .842}$ $= .36$, so the SE of the sum is $\sqrt{500} \times .36 = 8$, and the SE of the percent is $(8/500)\times 100\%$ $= 1.6\%$.

   (a) 15.8%, 1.6%.

   (b) The confidence interval is 15.8% $\pm$ 3.2%.

2. The sample is like 500 draws from a box with 25,000 tickets. The tickets are marked 0 (does not have a refrigerator) or 1 (has a refrigerator). We do not know how many tickets are marked 0 and how many are marked 1. We can approximate the proportion of 1's using the sample percentage: $(498/500)\times$ 100% $= 99.6\%$. We can approximate the SD of the box by $\sqrt{.996 * .004} = .063$, so the SE of the sum is $\sqrt{500} \times .063 = 1.41$, and the SE of the percent is $(1.41/500)\times 100\% = 0.3\%$.

   (a) 99.6%, 0.3%.

   (b) The confidence interval cannot be found because the box is so lopsided that the normal curve is not valid.

4. (a) The data are like 6000 draws from a box with 0's (for those students who didn't know that Chaucer wrote it), and 1's (for those students who did know that Chaucer wrote it.) The fraction of 1's in the box can be estimated by the fraction of 1's in the sample, which is 0.361. So, the SD of the box can be estimated as $\sqrt{0.361 \times 0.639} = 0.48$. The SE for the sum is $\sqrt{6000} \times 0.48 = 37$, and the SE for teh percentage is $(37/6000)\times$ 100% $= 0.6\%$. So the 95% confidence interval is 36.1% $\pm$ 1.2%.

   (b) 95.2% $\pm$ 0.6%.

11. Option (ii) is the best - for example, if the two percentage points is 2 SE's, then 95% of the estimates will be right to within two percentage points.

Chapter 22

3. The estimate is 9.0 million and the estimates SE is 0.1 million (the half sample method).

7. (a) This is a probability sample because it is based on an objective chance process.

   (b) It is not a simple random sample because only people from 2 districts will be chosen.

8. (a) This is a probability sample because it is based on an objective chance process.

   (b) It is not a simple random sample because in a simple random sample there would be a chance that both the 17th and 18th could be in the sample - here that cannot happen.

12. (a) True

    (b) True

    (c) False - the normal curve doesn't work because the data are too skewed. (The confidence interval goes negative, and the percentage cannot be negative).

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Homework 15 Solutions

Chapter 23

2. (a) True.
   (b) True.
   (c) False - it's a confidence interval.

3. The data are like 1000 draws from a box with 50,000 tickets. Each ticket shows the commute distance for the head of the household. The SD of the box is unknown, but we can estimate it using the SD of the sample, as 9.0 miles. The SE of the sum is $\sqrt{1000} \times 9.0 = 285$ miles. The SE of the average is $285/1000 = 0.3$ miles. The data do not follow the normal curve, but the average of 1000 draws will follow the normal curve, so we can obtain confidence intervals.
   (a) 8.7 miles, 0.3 miles.
   (b) 8.7 miles $\pm$ 0.6 miles.

6. (a) We cannot do it - we need to use the 2-sample method.
   (b) The SE of the average is approximately 1.

8. (a) True.
   (b) True.
   (c) Probably false - the data do not follow the normal curve.
   (d) False - 325 is the SE not the SD.
   (e) False - the normal curve is used for confidence intervals.

9. 1.7 $\pm$ 0.1.

10. (a) True.
    (b) False - there is no such thing as a CI for the sample average.
    (c) True.
    (d) False - that used the SE, and also, there must be a whole number of people.
    (e) False - if it did, there would be many households with a negative number of people.
    (f) True.

11. There is too much spread - the SE of the average is only 0.3, and for that histogram the SE is closer to 3.0.

12. This is not a confidence interval - there is no sample. At best, the "sample" is a sample of convenience, so we cannot find confidence intervals.

Stat 1040
Homework 16 Solutions

Chapter 24

1. (a) 81,411 inches; 6 inches (= $\sqrt{25}\times30$ / 25).
   (b) True.
   (c) False. We know the average of the 25 readings exactly.
   (d) False. Incorrect give or take number. Should be using the SD.
   (e) False. Same issue as part (d).
   (f) False. Confidence intervals are used to say something about the population based on the sample. We can't use them to say anything about a new sample.

2. (a) ... the probability histogram for the average of the draws follows the normal curve.
   (b) ... the histogram for the data doesn't follow the normal curve.

3. The SE is $\sqrt{2500}\times14$ / 2500 = 0.28 and the confidence interval is 299,744 $\pm$ 0.6.

4. We need the SD of the measurements.

6. Besides the fact that the Gaussian model doesn't apply here–think what a graph of average daily maximum temperature would look like, there would be "runs" of warm days and cold days, patterns that don't show up in draws made at random from a box–besides all that, we know the average exactly, and have no need to calculate a confidence interval.

7. Confidence interval is 78.1 $\pm$ 5.1 micrograms above 1 kilogram. Use the SD of the old data: 18 micrograms.

8. The data are like 100 draws made at random with replacement from a box. The average of the box is about 58 seconds, and the SD is about 2 seconds. The total execution time is like the sum of 100 draws from the box. This will be about 100×58 = 5800 seconds, give or take $\sqrt{100}\times2$ = 20 seconds or so.

9. (a) The weights of the sticks are drawn at random from a box; the box has an average of 4 ounces, and the SD is 0.05 ounces. The weight of a package is like the sum of 4 draws. The expected value is 4×4 = 16 ounces. The SE for the sum is $\sqrt{4}\times0.05$ = 0.1 ounces.
   (b) The total weight of 100 packages is like the sum of 400 draws form the box. The expected value is 400×4 = 1600 ounces = 100 pounds. The SE for the sum is $\sqrt{400}\times0.05$ = 1 ounce. The total weight will be 100 pounds, give or take 1 ounce or so. The range "100 pounds $\pm$ 2 ounces" is "expected value $\pm$ 2 SEs", so the chance is about 95%.

Stat 1040
Homework 17 Solutions

Chapter 26

1. (a) True.

   (b) False. It's the null that says the difference is due to chance.

2. The data are like 3800 draws made at random from a box containing 0's and 1's (1 stands for red), but with the number of each ticket unknown.

   (a) Null: the fractions of 1's in the box is 18/38. Alt: The fraction of 1's in the box is more than 18/38.

   (b) The expected number of reds (using the box from the null hypothesis) is 1800. The SD of the box (again using the null hypothesis) is nearly 0.5, so the SE is $\sqrt{3800} \times 0.5 =$ (about) 31. So z = (obs – exp)/SE = (1890 – 1800)/31 = (about) 2.9, and P = (about) 2/1000.

   (c) Yes.

3. Null hypothesis: the data are like 200 draws from the box containing three 1's and a single 0 (1 = blue and 0 = white). The expected number of blues is 150, with a SE of about 6. So z = (obs – exp)/SE = (142 – 150)/6 = (about) –1.3 and P = (about) 10%. This is marginal, could be chance.

5. The box has one ticket for each freshman at the university, showing how many hours per week that student spends at parties. So there are about 3000 tickets in the box. The data are like 100 draws from the box. The null hypothesis says that the average of the box is 7.5 hours. The alternative says that the average is less than 7.5 hours. The observed value for the sample average is 6.6 hours. The SD of the box is not known, but can be estimated from the data as 9 hours, which puts the estimate for the SE for the average at 0.9 hours. Then z = (obs – exp)/SE = (6.6 – 7.5)/0.9 = –1. The difference looks like chance.

6. (a) This is like tossing a coin 350 times and asking for the chance of getting 102 heads or fewer. The expected number of heads is 175, and the SE is 9. The chance is about 0.

   (b) 100 draws are made at random without replacement from a box with 102 1's (for the women) and 248 0's (for the men). The expected number of 1's is 29. If the draws are made with replacement, the SE for the number of 1's is $\sqrt{100} \times \sqrt{0.29 \times 0.71} =$ (about) 4.54. The correction factor is $\sqrt{(350-100)/(350-1)} =$ (about) 0.846. The SE for the number, when drawing without replacement, is $0.846 \times 4.54 =$ (about) 3.8. The chance is about 0.

   (c) Judge Ford was not choosing at random, he was excluding women.

8. There is one ticket in the box for each person in the county, age 18 and over. The ticket shows that person's educational level. The data are like 1000 draws from the box. Null: The average of the box is 13 years. Alt: The average of the box isn't 13 years. The expected value for the average of the draws (based on the null) is 13 years, give or take 0.16 years or so. (The 0.16 is the SE calculated from the SD of the data, 5 years. There is no reason to assume the SD of the nation equals the SD of the county, so the data provides a better estimate.) The observed value for the sample average is 14 years, so z = (obs – exp)/SE = (14 – 13)/0.16

= (about) 6, and P is about 0. This may be a rich, suburban county, where the educational level is above average.

9. Something is wrong. The EV for the sum is 20; the SE is 4. The average of 144 sums should be around 20; the SE for the average is 0.33. The observed value is 3.4 SEs away from the expected. That is too many SEs.

10. Null: the 3 Sunday numbers are like 3 draws made at random (without replacement) froma box containing all 25 numbers in the table. The average of these numbers is nearly 436, and their SD is just about 40. The EV for the average is 436, and the SE is 22 (using the correction factor). The 3 Sunday numbers average about 357, so z = (obs – exp)/SE = (357 – 436)/22 = (about) –3.6, and P = (about) 2/10,000. Many deliveries are induced, and some are surgical, so obstetricians really can influence the timing and avoid deliveries on Sunday (when they would rather be golfing).

12. (a) There are 59 pairs. The treatment animal has a heavier cortex in 52 of the 59 pairs. If the treatment has no effect, the number where the treatment animal has the heavier cortex is like the sum of 59 draws from a box containing one 0 and one 1. The expected value of the sum is $59 \times 0.5 = 29.5$, and the SE is $\sqrt{59} \times 0.5 = 3.84$. The observed number is 52, which is several SE's above the expected value, so the P-value is almost zero - there's virtually no chance of obtaining as many pairs as Rosenzweig did, or more, with the treatment animal having the heavier cortex. We conclude that the treatment did have an effect.

Stat 1040
Homework 18 Solutions

Chapter 27

1. No. If they were correct, the number of positives should be like the sum of 500 draws from a box with one "1" and one "0". The expected number of positives is 250 and the SE is $\sqrt{500} \times 0.5 = 11$. The observed number of positives is 2.4 SE's above what we would expect, so we conclude that more than 50% of the tickets in the box show positive numbers.

3. (a) There are two samples, so we need a 2-sample z-test.
   (b) There are two boxes. In the 1985 box, there is a ticket for each person in the population, marked "1" for those who would rate clergymen "very high or high" and marked "0" otherwise. The 1985 data are like 1000 draws from this box. The 1992 box is set up the same way, for 1992. The null hypothesis says that the percentage of "1"s in the 1985 box is the same as the percentage of "1"s in the 1992 box. The alternative hypothesis says that the percentage of "1"s in the 1992 box is smaller than that for the 1985 box.
   (c) The SD of the 1985 box is estimated as $\sqrt{0.67 \times 0.33} = 0.47$ and the SE of the percent is 1.5%. Similarly, the SE for the 1992 percent is 1.6%. The SE for the difference between these two percentages is $\sqrt{1.5^2 + 1.6^2} = 2.2\%$. The observed percentages are 67% (1985) and 54% (1992), so the z-value is $z = (67 - 54)/2.2 = 6$ and the P-value is almost zero. We conclude that the difference is real (we cannot say what the cause is).

7. (a) The SE for the treatment percent is 2.0% and for the control percent it's 4.0%. The SE for the difference between the two percents is 4.5%. The observed difference is 1.1%, so $z = 0.24$ and the P-value is large, so the observed difference could easily be due to chance. There is no evidence that income support reduces recidivism.
   (b) The SE for the treatment average is 0.7, for the control it's 1.4, and for the difference it's 1.6. The observed difference is -7.5 weeks, so $z = -7.5/1.6 = -4.7$ and the P-value is almost zero. We conclude that the difference is real – income support makes the released prisoners work less.

8. (a) The sample percentages are 47.8% and 4.4%. The standard errors for these percentages are 7.4% and 3.0%. The SE for the difference is 8%, so z=(47.8-4.4)/8 = 5.4. The P-value is almost zero, so we reject the null hypothesis and conclude that the difference is real - people are more likely to say they would do it than to actually do it.
   (b) The sample percentages are 30.4% and 4.4%. The standard errors are 6.8% and 3.0%. The standard error of the difference is 7.4%, so z=(30.4-4.4)/7.4 = 3.5 and the P-value is very small. We reject the null hypothesis and conclude that people are more likely to volunteer if they have been first asked to predict whether they would.
   (c) These are the same people, so a 2-sample z-test is not valid (we need *independent* simple random samples).

10. These are children from the same 400 families and the sampling was not random, so a 2-sample z-test is not valid (we need *independent* simple random samples).

Stat 1040
Homework 19 Solutions

Chapter 28

1. (a) (i)
   (b) (iii)

2. The observed counts are 1, 10, 16, 35 and the expected counts are 28.4% of 62 = 17.6, 48.5% of 62 = 30.1, 11.9% of 62 = 7.4 and 11.2% of 62 = 6.9. The χ^2 value is 150 on (4-1)=3 degrees of freedom, so option (ii) is correct. Judges prefer educated jurors.

3. This is a test for independence. The expected counts are:

654.1	109.3	132.6
67.9	11.3	13.8
62.0	10.4	12.6

   The χ^2 value is about 30 on (3-1)×(3-1) = 4 degrees of freedom, so the P-value is very small and we reject the null. We conclude that this is not chance variation - there are differences in employment levels for different marital status.

7. The expected counts are 93.5, 187, 93.5. The χ^2 value is about 0.2, on 2 degrees of freedom. The P-value is about 90%, so this is a good fit.

Chapter 29

4. We would expect 5% of them to be found guilty - this is data snooping.

5. The question makes no sense - there is no box model - there are no samples, nor any randomization.

7. The difference is practically significant, but statistical significance makes no sense because the data are for the whole population.

9. (a) The question makes sense and the difference in attitudes is important.
   (b) The question makes sense because they took probability samples. However, we cannot answer it because they are cluster samples.
   (c) The two percentages are 48% and 25%. The two standard errors are 1.6% and 1.4%. The SE for the difference is 2.1% so z=(48-25)/2.2 = 10 and the P-value is tiny. We reject the null and conclude that attitudes changed.